

Zeros of Linear Functions from Graphs

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The **zero** of a linear function is the value of x
that makes $y = 0$.

Zero and x-Intercept

The **zero** is also called the **x-intercept** because it is the point where the graph crosses the x -axis.

Finding the Zero from a Graph

To determine the zero from a graph, locate the point where the line intersects the x -axis.

Coordinates of the x-Intercept

The coordinates of the x-intercept are written as $(a, 0)$, where a is the zero of the function.

Example of a Zero

If the graph crosses the x -axis at $(5, 0)$, then the zero of the function is $x = 5$.

Number of Zeros

Every non-horizontal linear function has exactly one **zero**.

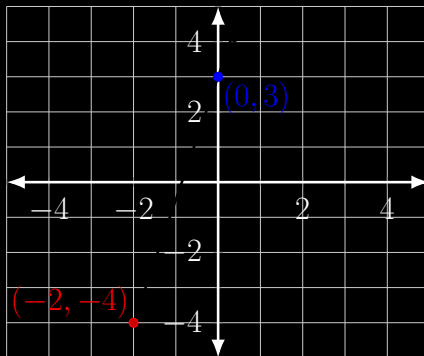
Possible Values of the Zero

The zero can be **positive**, **negative**, or **zero** depending on where the graph intersects the x -axis.

Practice Exercises

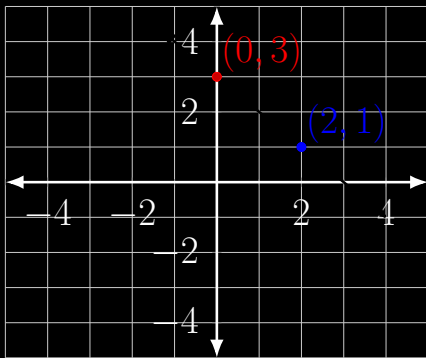
Practice 1

For the graph, determine the **zero** of the function.



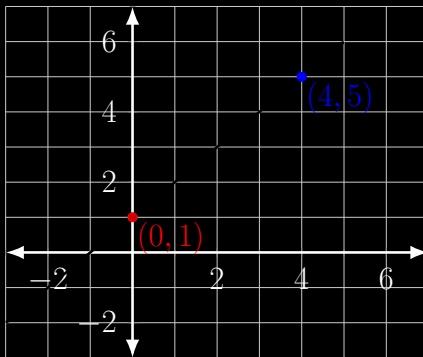
Practice 2

For the graph, determine the **zero** of the function.



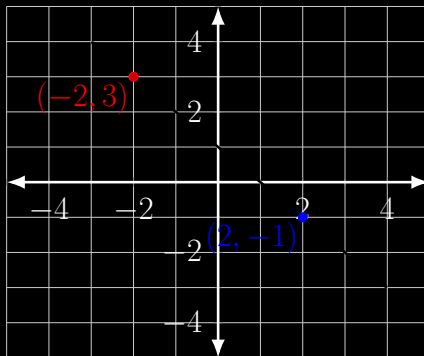
Practice 3

For the graph, determine the **zero** of the function.



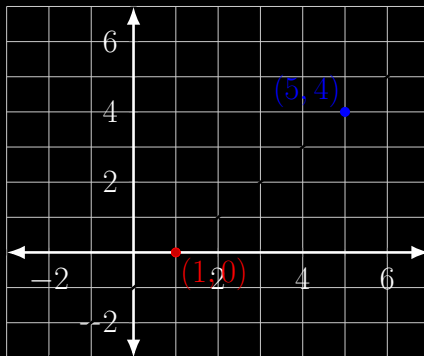
Practice 4

For the graph, determine the **zero** of the function.



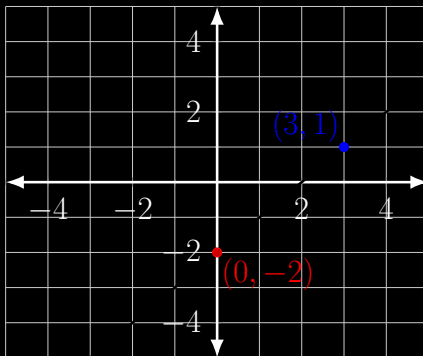
Practice 5

For the graph, determine the **zero** of the function.



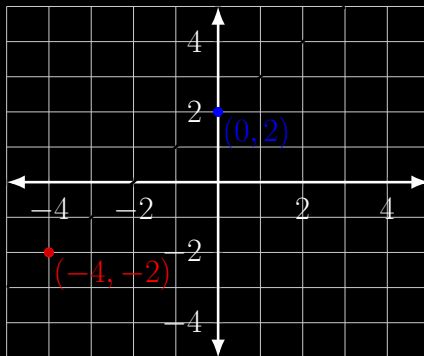
Practice 6

For the graph, determine the **zero** of the function.



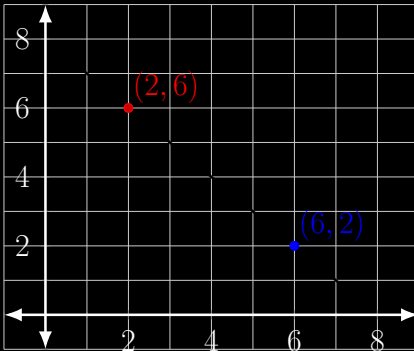
Practice 7

For the graph, determine the **zero** of the function.



Practice 8

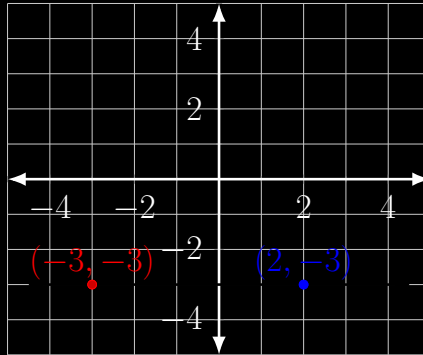
For the graph, determine the **zero** of the function.



Activities

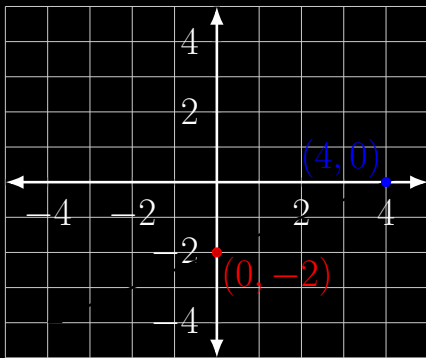
Activity 1

For the graph, determine the **zero** of the function.



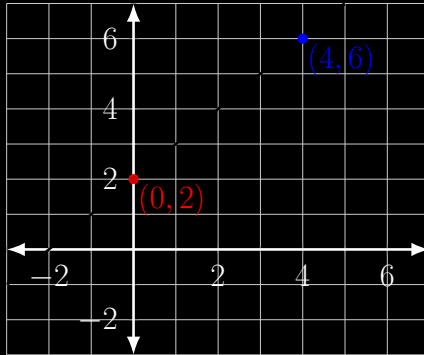
Activity 2

For the graph, determine the **zero** of the function.



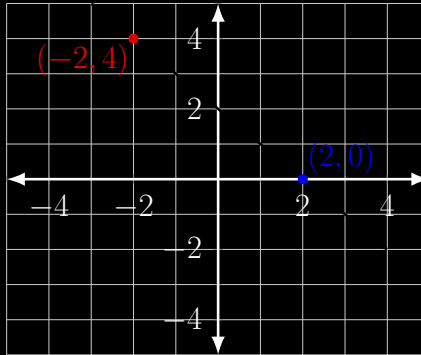
Activity 3

For the graph, determine the **zero** of the function.



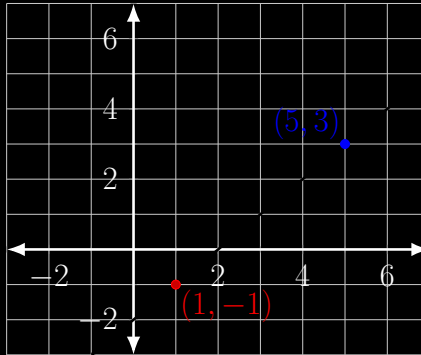
Activity 4

For the graph, determine the **zero** of the function.



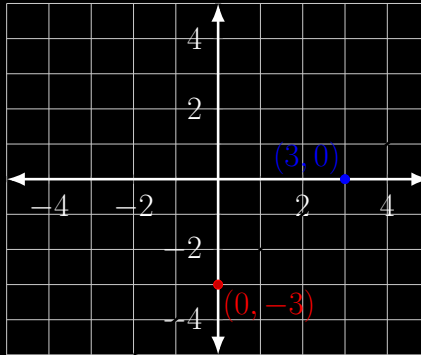
Activity 5

For the graph, determine the **zero** of the function.



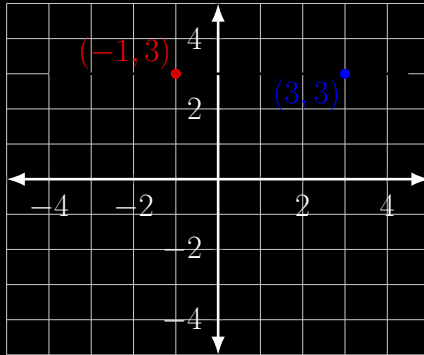
Activity 6

For the graph, determine the **zero** of the function.



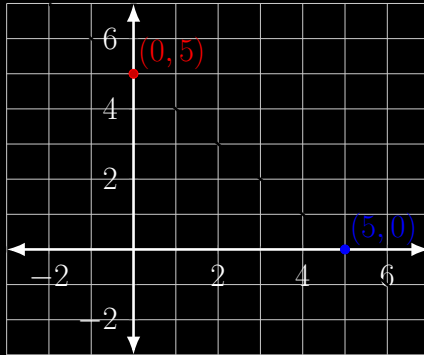
Activity 7

For the graph, determine the **zero** of the function.



Activity 8

For the graph, determine the **zero** of the function.



Thank you!